Assignment 4

Q1. Load load_breast_cancer() built in dataset.

Q2. convert dataset to pandas dataframe.

Q3. Append dataframe containing tumor features with diagnostic outcomes.

This labels will be used for supervised learning.

Q4.Find relation in diagnosis by using mean.

Q5. create to dataframes - one for positive, one for negative(diagnosis).

Q6. Visualize tumor characteristics for positive and negatives diagnoses

Q7. Create 'for loop' to enerate though tumor features and compare with

histograms(20:-1)

Q8. Find Visualization of relationships between features and diagnoses.

Q9. Split data into testing and training set. Use 80% for training

Q10. normalize features to account for feature scaling

Q11. Apply DT, NB, KNN Logistic Regression and SVm to find accuracy, F1-Score and Recall.

Q12. Further apply the PCA to do the same exercise in Q11.

Q13. Do the Comprative study with PCA and WITHout PCA.

Question 1.1

Q1. Apply Mc-Culloch Pits model for the following Boolean operations:

Plot the decision boundaries for 2 - dimensional and 3-dimensional.

Implement AND with 3-variables and threshold 3.

Implement OR with 3-variables and threshold 1.

Implement X 1and !X2 with 2-variables and threshold 1.

Implement NOR with 2-variables and threshold 0.

Implement NOT with 2-variables and threshold 0.

Implement OR with 2-variables and threshold 1.

Implement AND with 2-variables and threshold 2.

Implement OR with 3-variables and threshold 1.